

Aperture Intelligence

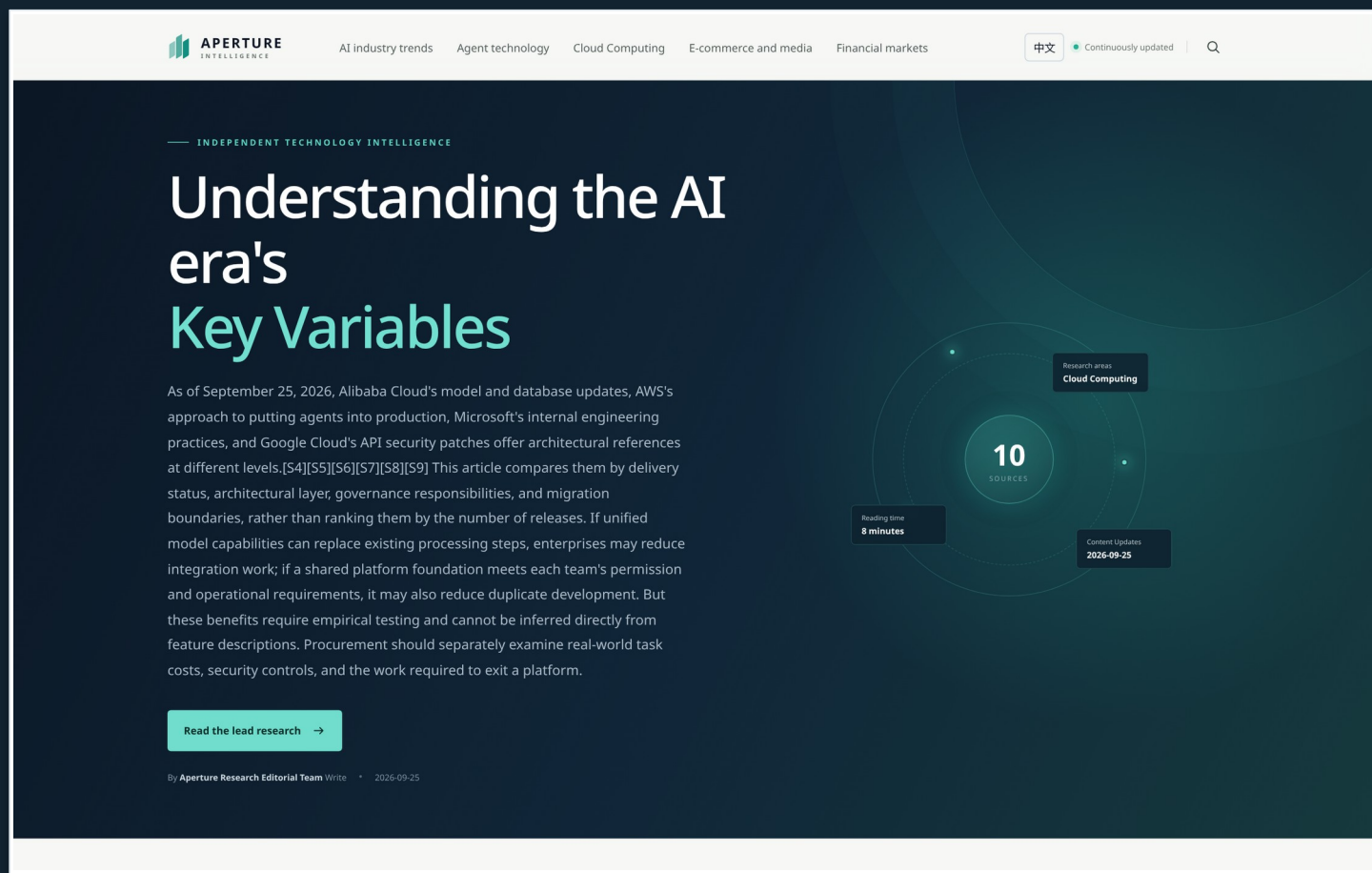
From public evidence
to reviewed research.

English-first publishing.

Permanent public URLs.

Programmatic agent access.

Technical overview • 25 September 2026



One system, three useful outcomes

01

Research with evidence

Collect public material, save source evidence, and review generated conclusions.

02

Publish for discovery

Serve readable English and Chinese editions at stable, crawlable URLs.

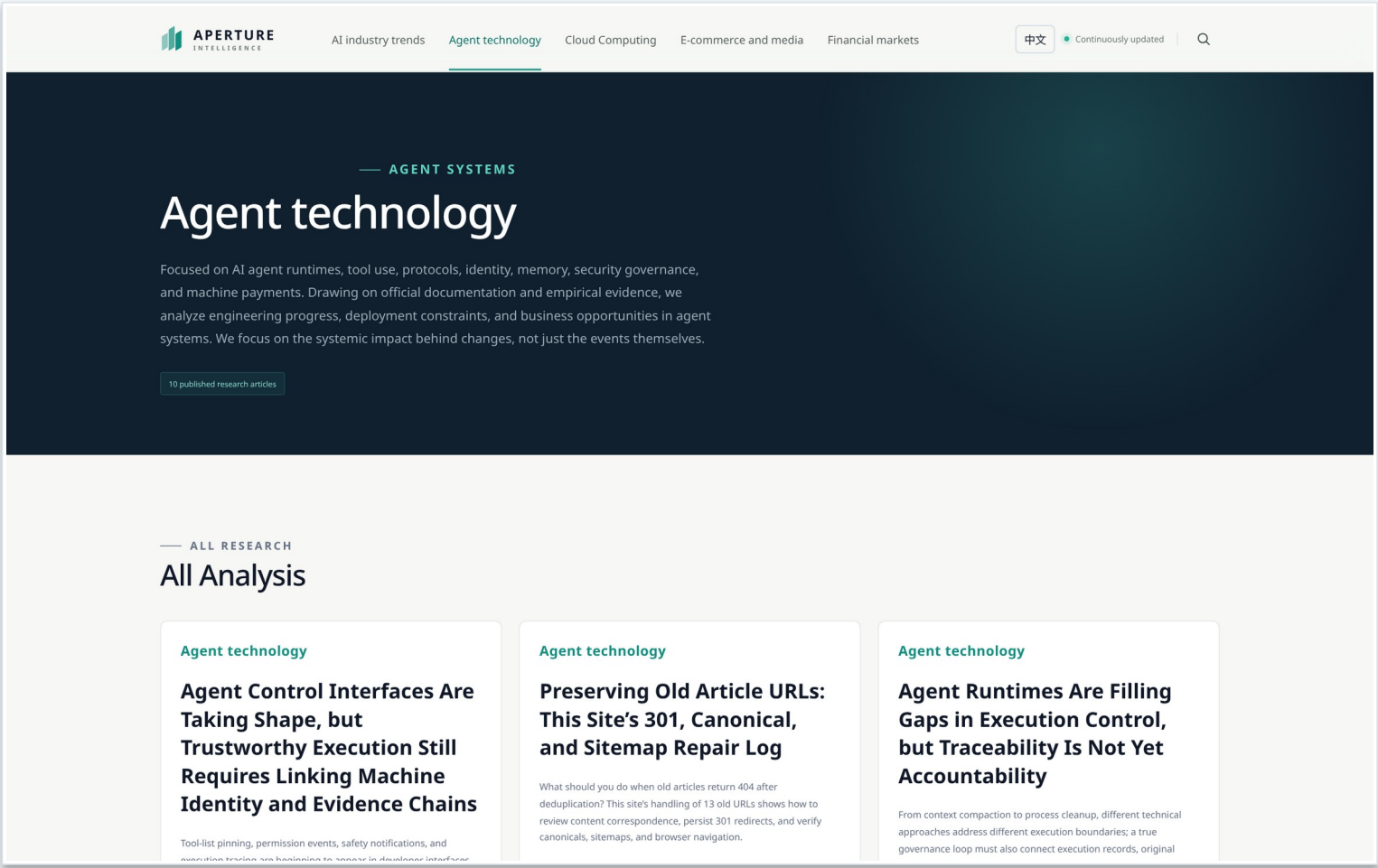
03

Serve people and agents

Offer public HTML, open JSON and an x402 payment-protocol demonstration.

Built for engineers exploring the full research-to-publication lifecycle.

A browsable research publication



Five research areas

- AI industry
- Agent technology
- Cloud computing
- Commerce and media
- Financial markets

Complete categories,
server-rendered links.

English by default, Chinese alongside

One article, two editions

EN `/article/{slug}`

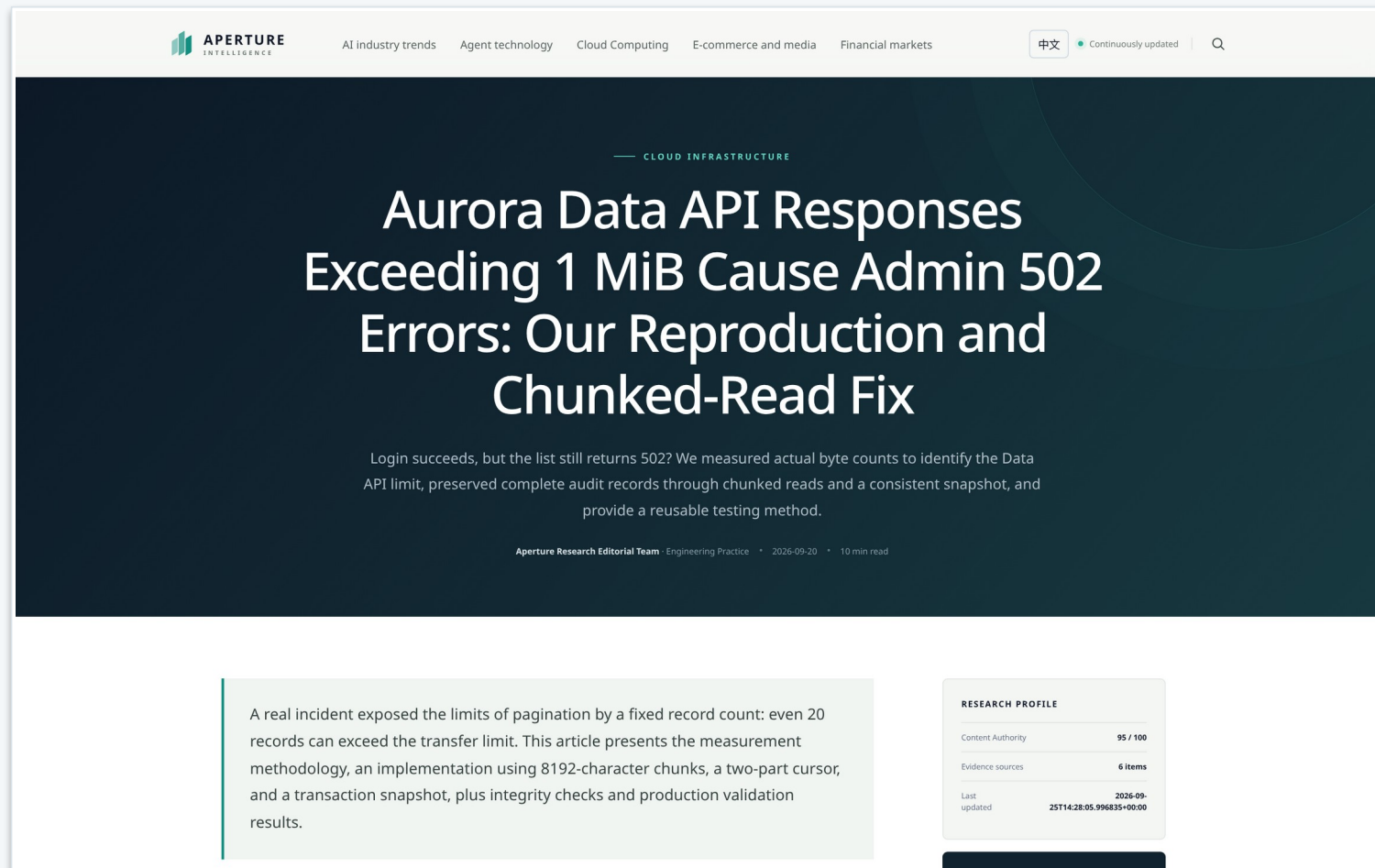
ZH `/zh/article/{slug}`

Full reviewed translations.

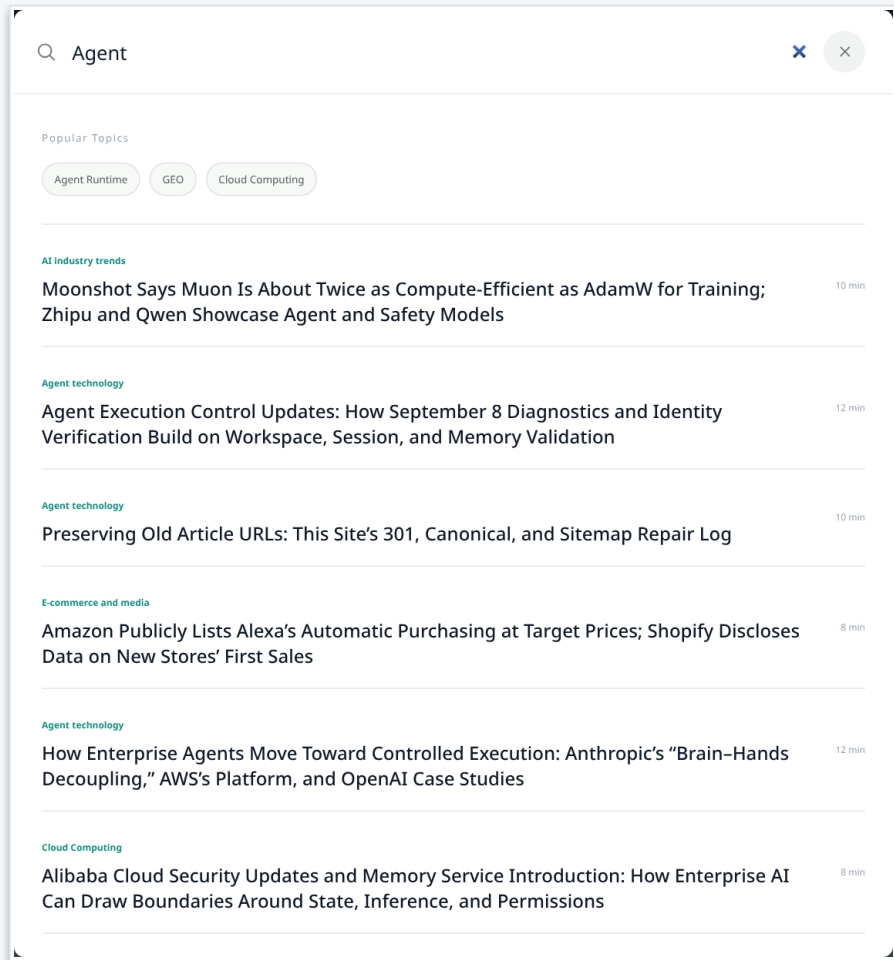
Page-preserving switching.

Canonical + hreflang.

Bilingual sitemap and RSS.



Discovery continues inside the site



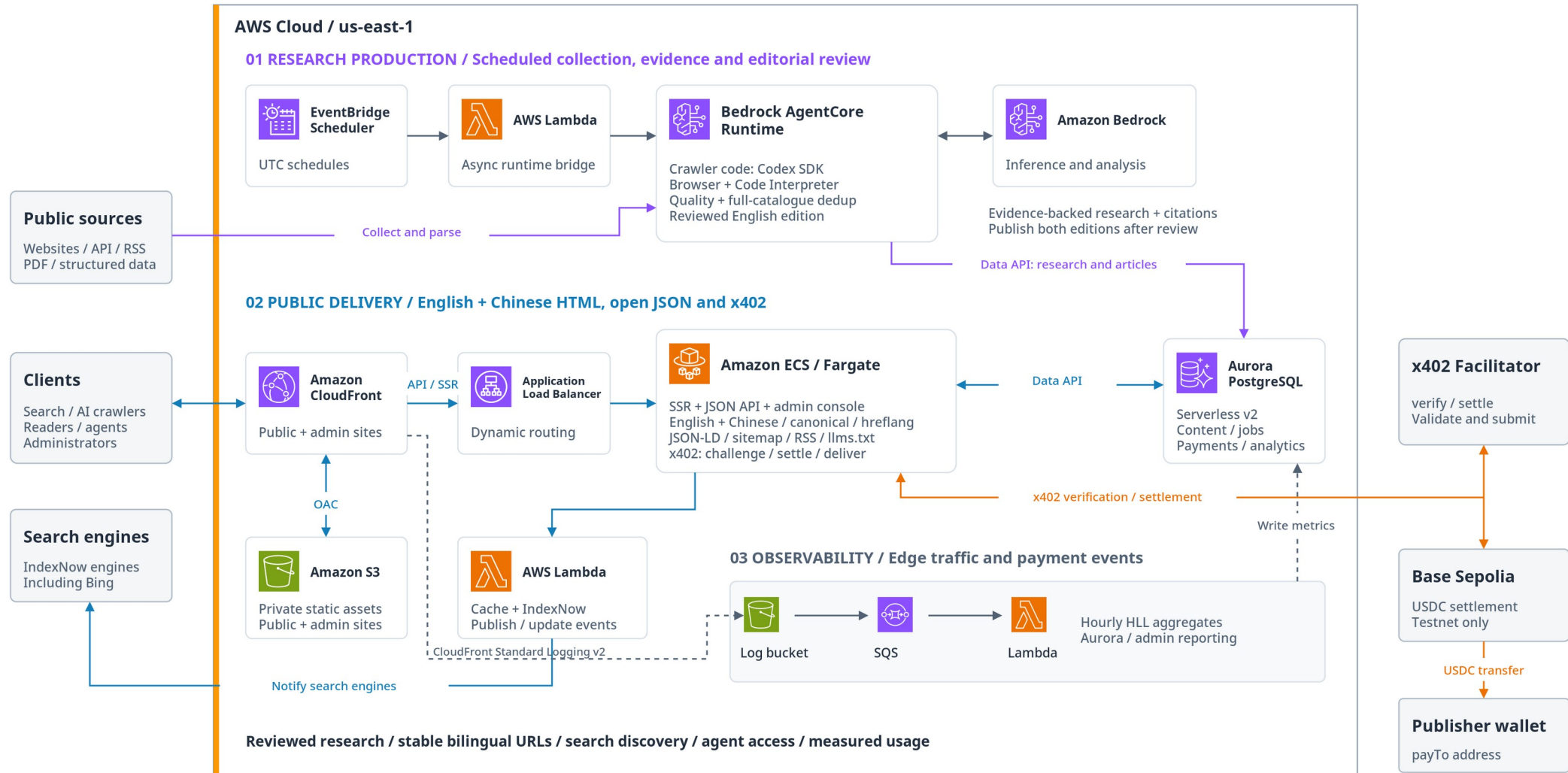
Search the reviewed catalogue

Search results use the selected language.
Categories expose every published article.
Related reading and RSS offer a next step.
Direct visits and client navigation share SSR.

Verified discovery paths. No ranking guarantee.

Aperture | Research, bilingual publishing and agent access

Evidence-based research, permanent public URLs and x402 protocol-based delivery



Public research remains accessible. IndexNow acceptance does not confirm indexing. Payments require a compatible wallet and spending authorization.

Implementation: milan9527/geo / docs/architecture.md / backend/x402_payment.py / Snapshot: 25 Sep 2026

Research runs asynchronously

1

Schedule and dispatch

EventBridge schedules a Lambda bridge. The bridge waits for runtime acceptance.

2

Collect and analyze

The runtime selects sources, generates crawler code and retrieves evidence with managed tools.

3

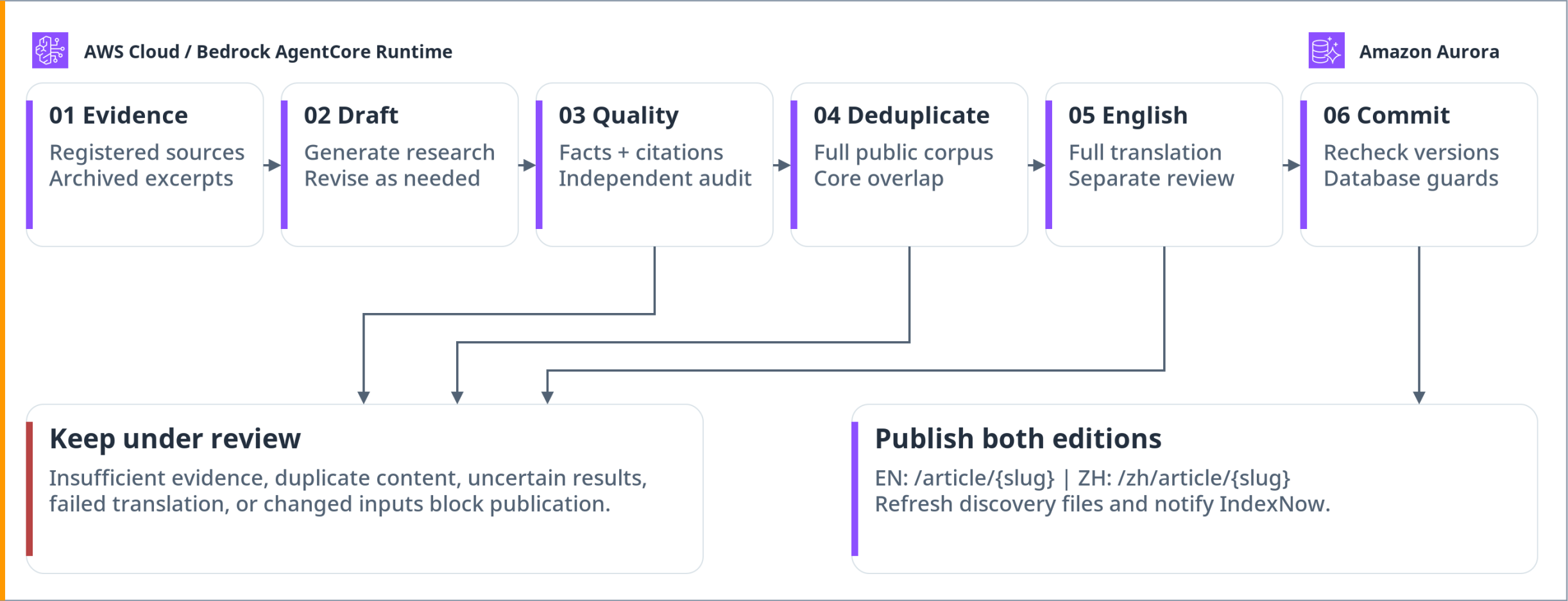
Record and review

Aurora stores job state, evidence, research output and publication decisions.

Codex SDK generates crawler code; Browser and Code Interpreter perform retrieval and execution.

Review before every new publication

Scheduled and manual reviews use the same quality, deduplication and bilingual safeguards



Existing published URLs remain protected. New content cannot replace or unpublish an older page.

New content must not erase old URLs

41

published articles

104

bilingual public URLs

19

legacy redirect mappings

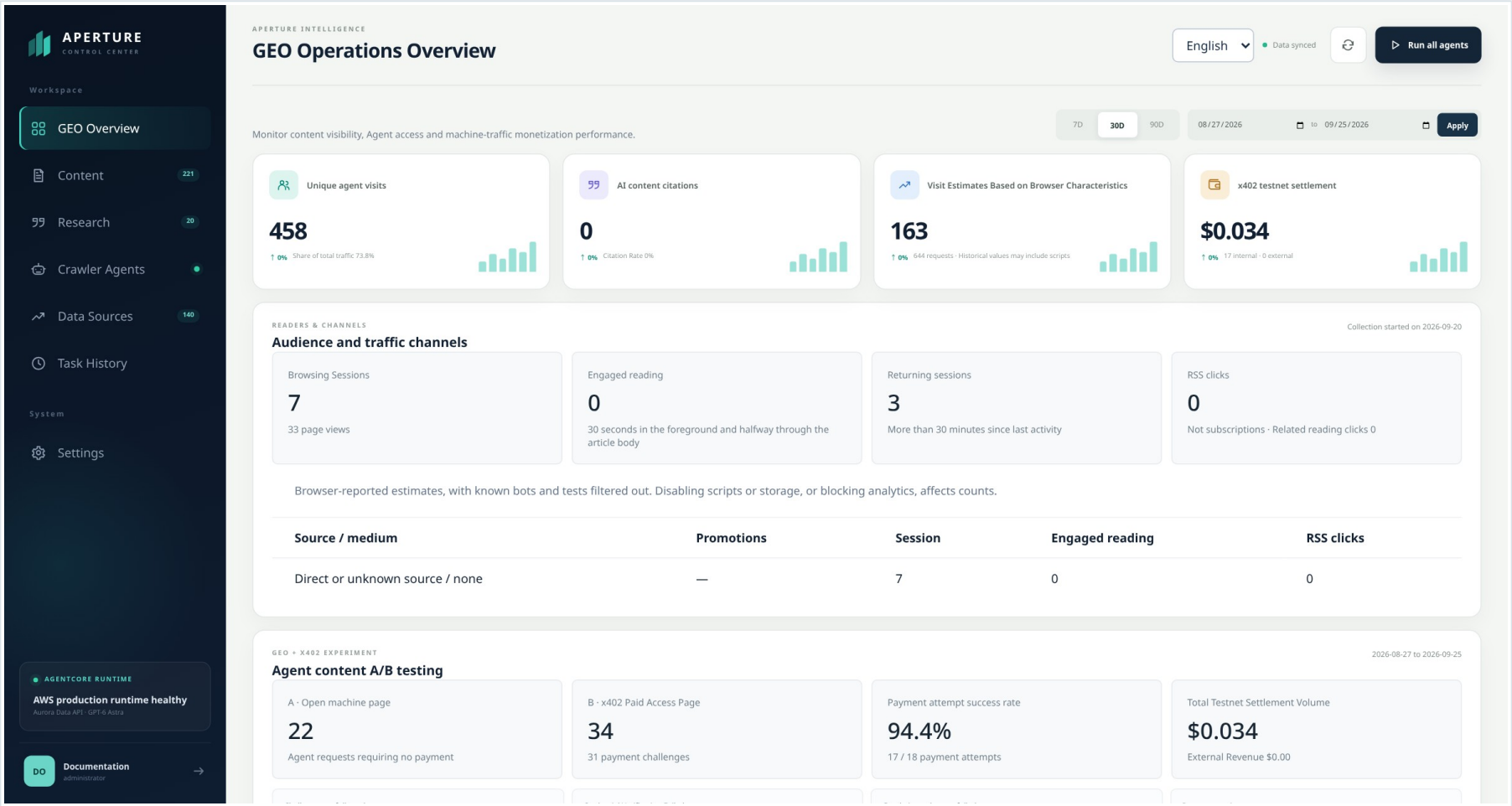
Published status and slugs are protected by the database.

Reviewed repairs use a specific 301 or restore the original URL.

New duplicates remain under review; existing pages stay available.

Snapshot: 25 Sep 2026 • Bing accepted the update; acceptance does not confirm indexing.

An English console for daily operations



Traffic estimates, reader attribution, research activity and x402 testnet events in one view.

Published content uses reviewed English editions

PUBLISHED CATALOGUE

APERTURE
INTELLIGENCE

Workspace

GEO Overview

Content221

Research20

Crawler Agents

Data Sources140

Task History

System

Settings

AGENCYCORE RUNTIME

AWS production runtime healthy

Documentation

APERTURE INTELLIGENCE

Content

EnglishData syncedRun all agents

Expert Content Library

Published pages remain online; new drafts are published automatically after passing evidence review and full-text deduplication.

Export contentCreate Research

Search titles or authors...Published

Selected 0 articlesMove to reviewDelete Unpublished Draft

☐	Content	Category	Status	GEO Authority	AI citations	Agent visits	Updated at	Actions
☐	Alibaba Cloud Updates Multimodal Models; Enterprise AI Architectures ... Aperture Research Editorial Team · @cloud-research-20260925-0415-1353	Cloud Computing	Published	89 / 100	10	Open	1 hours ago	Keep Public
☐	Amazon Seller Tool Updates and Shopify's Partner Strategy: Operational... Aperture Research Editorial Team · @amazon-research-20260923-0116-1307	E-commerce and media	Published	80 / 100	10	Open	1 hours ago	Keep Public
☐	Agent Control Interfaces Are Taking Shape, but Trustworthy Execution ... Aperture Research Editorial Team · @agent-research-20260923-0312-1309	Agent technology	Published	85 / 100	10	Open	1 hours ago	Keep Public
☐	Anthropic Emphasizes Lower Model Running Costs, OpenAI Improves C... Aperture Research Editorial Team · @ai-research-20260923-0219-1266	AI industry trends	Published	85 / 100	10	Open	1 hours ago	Keep Public
☐	From Cloud Access to Agent Memory: Delivery Paths and Conditions for ... Aperture Research Editorial Team · @ai-research-20260922-0215-1360	AI industry trends	Published	88 / 100	10	Open	1 hours ago	Keep Public
☐	Kimi Launches Hosted Agents, Zhipu Offers a Companion Runtime Fram... Aperture Research Editorial Team · @ai-research-20260921-0214-1254	AI industry trends	Published	88 / 100	10	Open	1 hours ago	Keep Public
☐	Preserving Old Article URLs: This Site's 301, Canonical, and Sitemap Rep... Aperture Research Editorial Team · @open-source-article-urls-301-canonical-sitemap	Agent technology	Published	94 / 100	0	Open	1 hours ago	Keep Public
☐	Aurora Data API Responses Exceeding 1 MiB Cause Admin 502 Errors: O... Aperture Research Editorial Team · @aurora-data-api-1mb-502-chunked-read	Cloud Computing	Published	95 / 100	0	Open	1 hours ago	Keep Public
☐	Stripe Discusses Pricing Preparations for Agent Buyers; Business Impac... Aperture Research Editorial Team · @stripe-research-20260922-0004-600	E-commerce and media	Published	91 / 100	14	Open	1 hours ago	Keep Public
☐	AWS, Baidu, and a Compile-by-Training Case Show How AI Tasks Are Org... Aperture Research Editorial Team · @ai-research-20260920-0410-804	AI industry trends	Published	90 / 100	10	Open	1 hours ago	Keep Public
☐	AI Capital Expenditure and Tech Stock Returns: Beyond Financing, Reve... Aperture Research Editorial Team · @finance-research-20260917-0212-1232	Financial markets	Published	92 / 100	15	Open	1 hours ago	Keep Public
☐	Alibaba Cloud Security Updates and Memory Service Introduction: How ... Aperture Research Editorial Team · @cloud-research-20260923-0909-170	Cloud Computing	Published	94 / 100	11	Open	1 hours ago	Keep Public
☐	From Snowflake Shared Compute to AWS Tenant Isolation: How Enterpr... Aperture Research Editorial Team · @aws-research-20260923-0909-170	Cloud Computing	Published	88 / 100	10	Open	1 hours ago	Keep Public

REVIEWED ARTICLE DETAIL

Cloud Computing · Published

Aurora Data API Responses Exceeding 1 MiB Cause Admin 502 Errors: Our Reproduction and Chunked-Read Fix

Aperture Research Editorial Team · Engineering PracticeAuthority 95/1000 citations2026-09-25 Update

Login succeeds, but the list still returns 502? We measured actual byte counts to identify the Data API limit, preserved complete audit records through chunked reads and a consistent snapshot, and provide a reusable testing method.

A real incident exposed the limits of pagination by a fixed record count: even 20 records can exceed the transfer limit. This article presents the measurement methodology, an implementation using 8192-character chunks, a two-part cursor, and a transaction snapshot, plus integrity checks and production validation results.

Why Does the Research List Still Return 502 After a Successful Login?

On September 14, 2026, login to the Aperture admin interface succeeded, but the research list endpoint returned 502, and the page reported that data had failed to load. Of the ten initialization requests, the research list request was the one that failed; the previously fixed statistics endpoint was working normally. This distinction matters: account authentication and research queries follow different paths, and logging in again cannot resolve a query response that exceeds the size limit.[S1]

Production logs traced the problem to the research list query and Aurora Data API's response size limit. The query JSON for the 20 most recent complete research records totaled 1,094,895 bytes, with growing audit data for database-wide deduplication increasing the size of individual records. At the time of the failure, the amount of data transferred could continue to grow even without an increase in the number of list entries.[S1]

This article reviews that incident, with code and tests pinned to the repository version publicly available on September 14. We do not treat latency measurements from a single fix as a performance guarantee for other environments, nor do we attribute all backend 502 errors to the Data API. Diagnosis should still begin with the specific failed request and its upstream error.

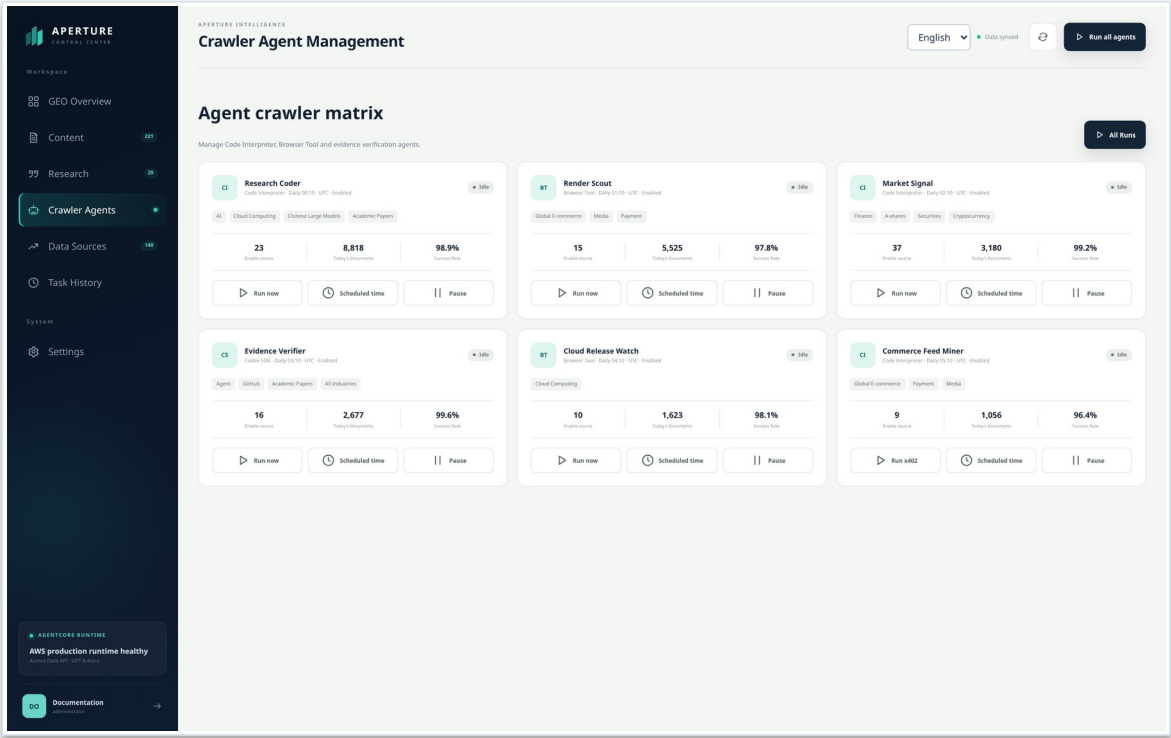
A Limited Row Count Does Not Mean a Limited Byte Count

AWS documentation specifies a 1 MiB result-set limit for the error Database response exceeded size limit. One MiB equals 1,048,576 bytes, and our query JSON at the time exceeded that amount. This byte count was a measurement of the query result, not the size of the browser page or the length of a single article's body.[S1][S4]

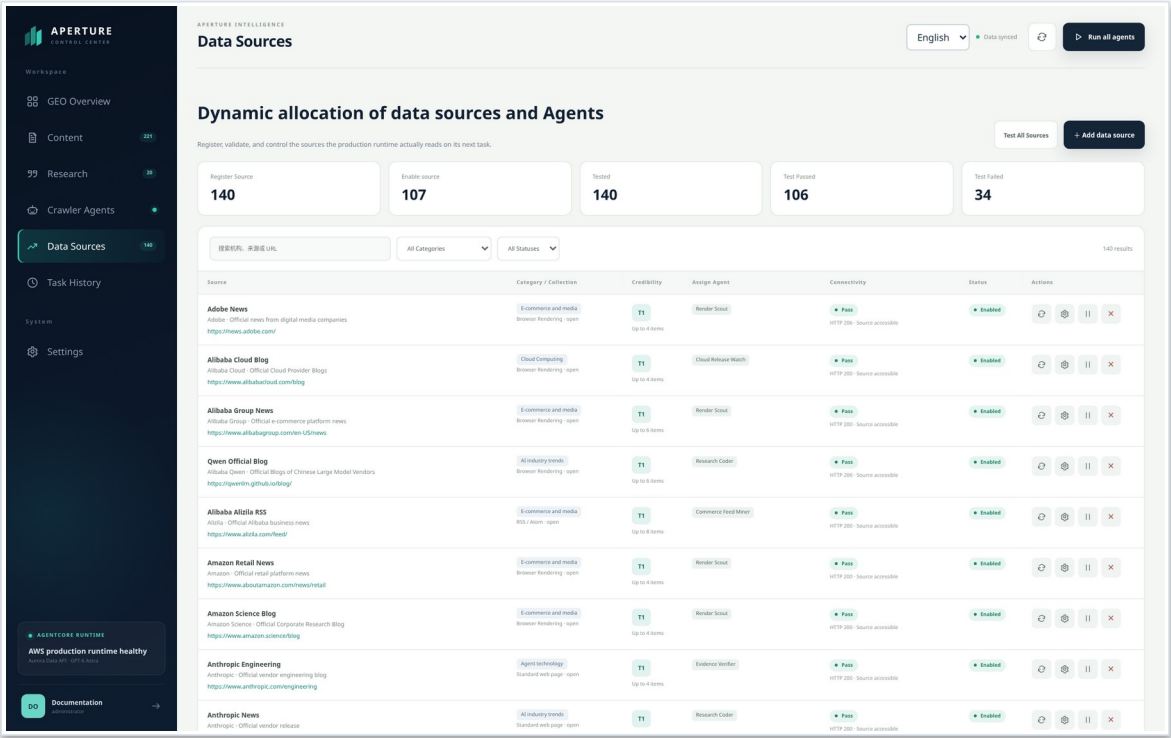
AWS recommends keeping the data returned by each Data API call within the limit, using multiple ExecuteStatement calls with LIMIT to retrieve results when necessary. Ordinary row-based pagination can reduce responses for many lists, but if a row contains a large audit JSON value, even LIMIT 1 cannot guarantee that a single response stays within the limit. AWS also separately documents a 64 KB per-row limit, so both the individual row size and the total response size

Original drafts and source evidence retain their original language and audit history.

Sources and schedules remain inspectable



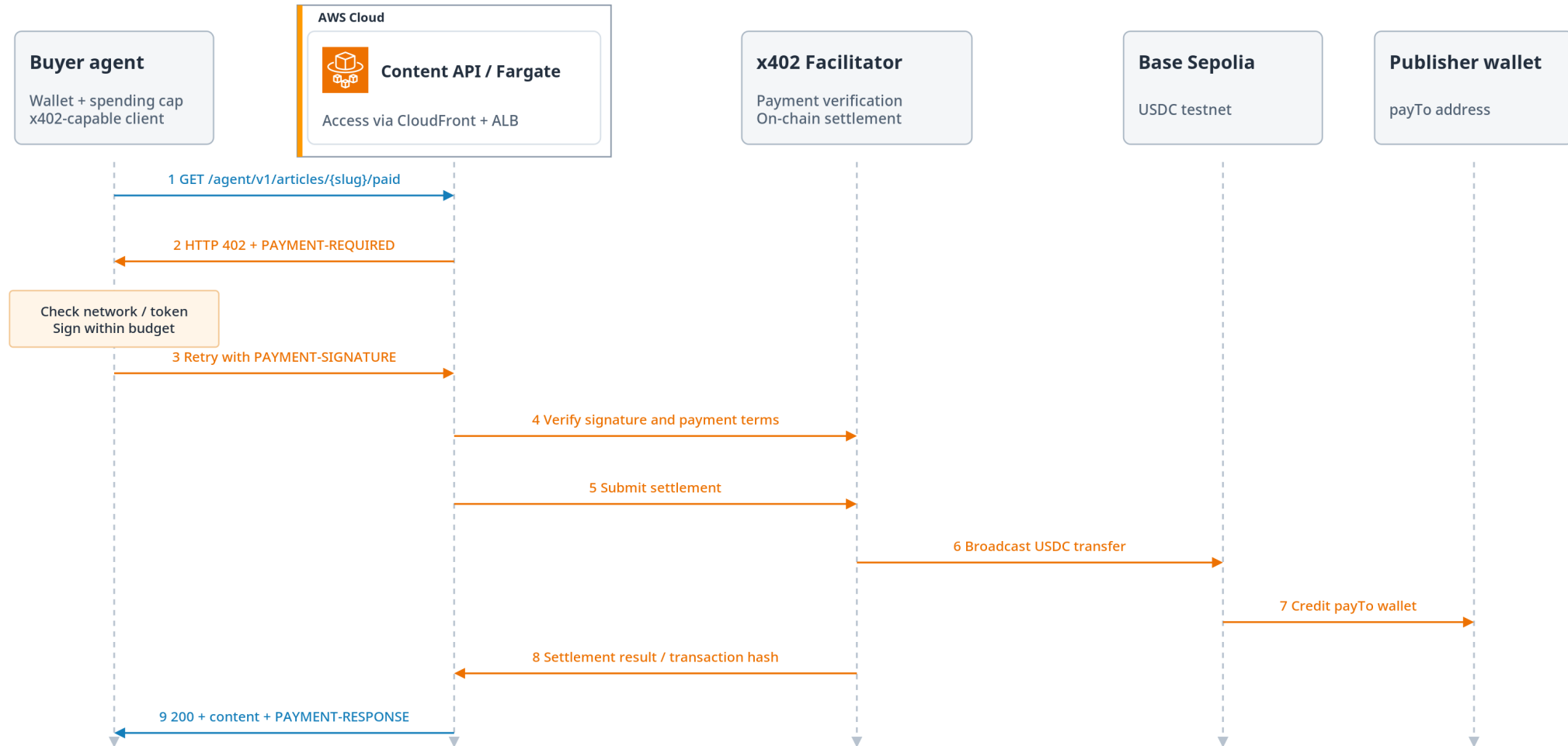
Crawler state, UTC schedules and recorded job outcomes.



Registered sources, assignments and connection-test workflows.

x402 | Agent payment and content delivery

Wallet authorization / HTTP 402 challenge / signed retry / verification and settlement / content delivery



MEASURE THE SEQUENCE / Request, challenge, signature, settlement and delivery

The API records payment and delivery events. Admin reporting combines these with aggregated CloudFront traffic.

Base Sepolia testnet USDC, default 0.002 USDC. Public research remains open; payment applies to the dedicated machine endpoint.

Measure observed behavior and expose limits

Two measurement sources

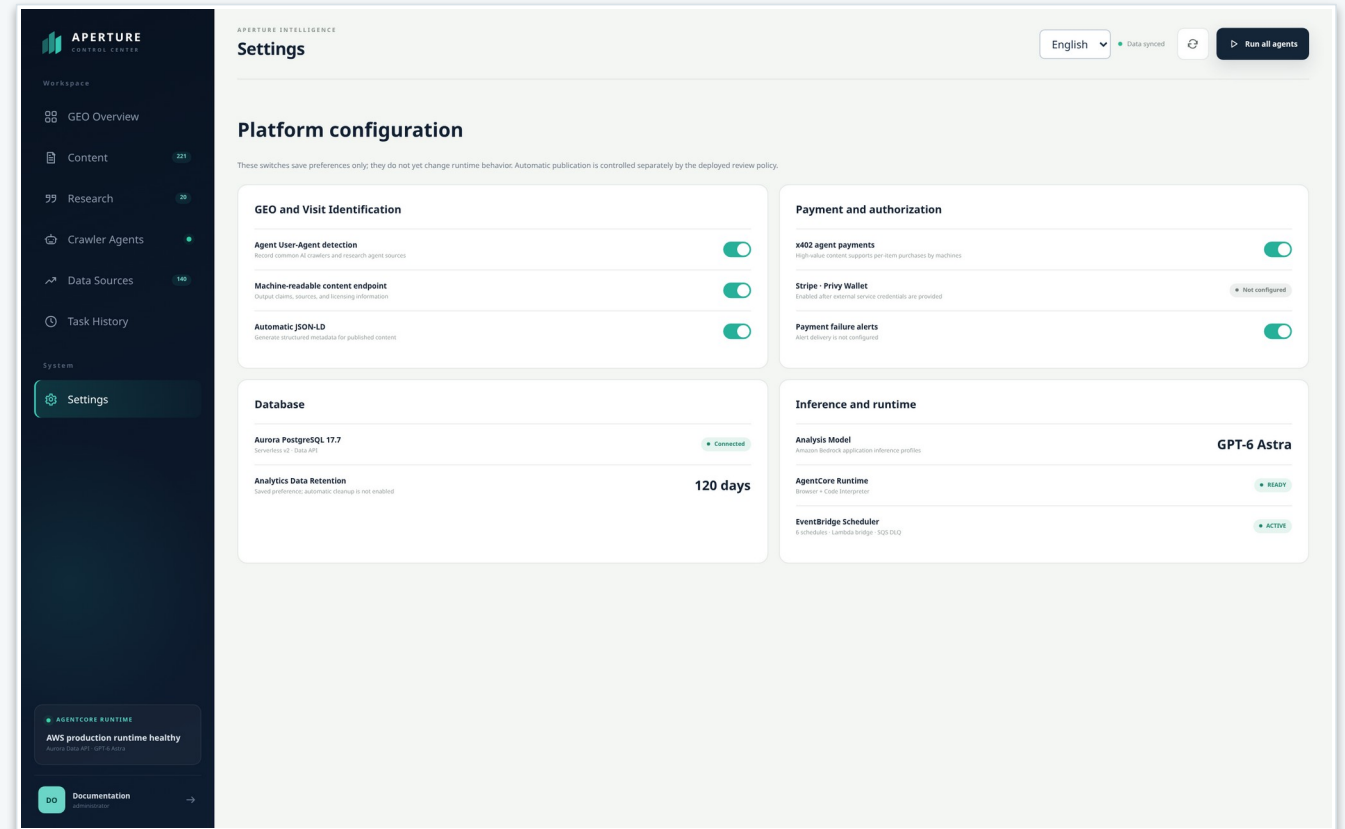
Edge logs: requests and visitor estimates.

Browser events: sessions, reading,
return visits and referral channels.

Exclude diagnostics from growth.

RSS clicks do not prove subscription.

Testnet payments are not revenue.



Preference-only controls are labeled in the actual UI.

A small application with explicit boundaries

frontend/public + frontend/admin	Reader experience and operations console
backend	SSR, APIs, auth, locales, persistence and x402 seller
aws_runtime + agent_runtime	Research orchestration, tools and publication gates
aws_scheduler + aws_indexing_notifier	Asynchronous invocation and discovery notifications
aws_traffic_aggregator	Edge-log classification and hourly aggregates
scripts + infrastructure/aws	Release workflows, tests, maintenance and IAM examples

Large Data API responses use bounded reads; review history and evidence remain intact.

Run the application locally

Python 3.10+ • Docker Compose

```
git clone https://github.com/milan9527/geo.git
cd geo
./scripts/start.sh
```

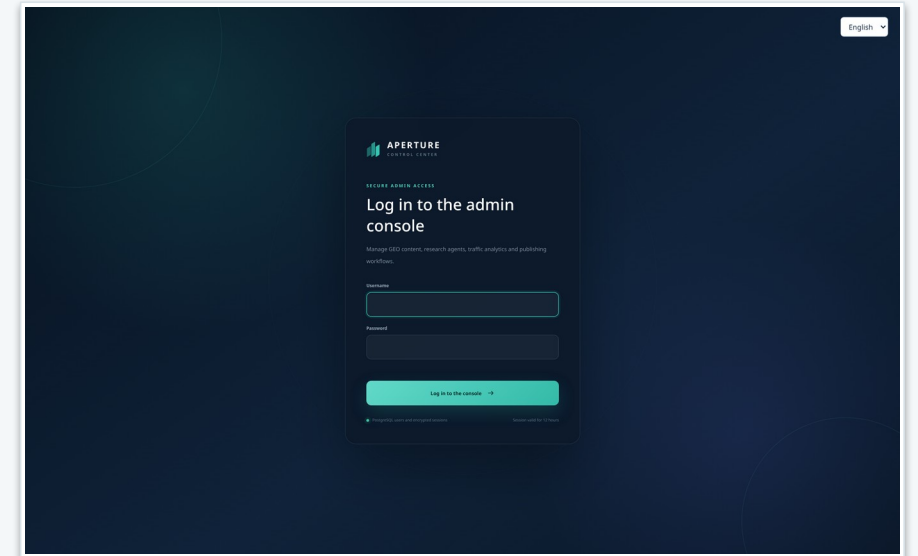
Public :4173 / Admin :4174 / API :8000

Create an admin with scripts/create_admin_user.py.

The command prompts for a password.

Node.js 24+ is needed for the crawler worker build.

See docs/deployment.md for configuration.



Actual English production login screen

Release into an existing AWS environment

Export `.env.aws` and `.env.deploy.aws` with your resource settings.

Provision the required AWS services before running release scripts.

```
./scripts/deploy_web_ecs.sh  
./scripts/deploy_agent_runtime.sh --auto-publish  
./scripts/deploy_scheduler_bridge.sh
```

Build and scan images → update services → upload assets → invalidate caches.

Wait for deployment and cache completion, then verify live bilingual routes.

Keep isolated tests separate from the production database.

Verified behavior, with evidence you can inspect

106

automated tests passed

312

crawler page checks passed

11

real English UI captures

**Both languages. Publication gates. Navigation and admin workflows.
Legacy redirects. Payment responses. Traffic attribution.**

Explore the site aperture.zhangwangshu.com

Read the code github.com/milan9527/geo

Reproduce checks docs/functional-verification.md